

Every Path a File Takes: An Annotated Anatomy of the OrbitBrief Evidence Pipeline

*A complete, graded, model-annotated account of the flow board, its downstream stages,
its learning loop, and the audit campaign that certified it*

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Abstract. This paper gives an exhaustive account of the OrbitBrief evidence pipeline — both of its stages — as depicted by its flow board (Figures 1–3). We proceed in the board's reading order — the evidence ladder that routes raw bytes, the five reader families and their interior routing, the convergence into segments and typed evidence atoms, the coverage floors and the recorded abstain — then continue past the board's edge into the stages atoms flow into: the evidence graph, the packetizer, phase-2 reconciliation, the cross-repository envelope contract, and brief generation. Each region receives a full functional description, an inventory of the machine-learned models operating within it (and a defense of the places where models are deliberately absent), a letter grade for its current state, and an argument for its novelty. We answer precisely where computation resides: every deployed component is CPU-only on Azure Container Apps; the only GPU in the system's life is a rented A100 used exclusively for training, fed from datasets staged on Azure blob storage; the Mac Studio on the office network serves optional embedding inference and the system is engineered to tolerate its absence. We document the 2026-08 seam-audit campaign — six audits, twenty-two defects, every one at a representation boundary and none inside rule logic — and the correction loop that turns a project manager's two-tap fix into both an instant, no-retrain behavior change and an accumulating training corpus. The system's present state is summarized without euphemism: fully audited, fully deployed, receipt-verified, and waiting — with zero verified human gold labels — for its first real correction. A glossary and four appendices are included so that no term, number, or claim in the paper requires outside context.

Keywords: document parsing · evidence routing · authority lattice · neuro-symbolic systems · abstention · invariance testing · human-in-the-loop learning · provenance

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1 Introduction and reading guide

OrbitBrief converts an unsorted folder of deal documents — quotes, statements of work, email threads, meeting transcripts, site rosters, floor-plan PDFs — into a project-manager-ready brief with receipts. It is built as a two-stage pipeline in two repositories: **parser-os** (stage one) parses artifacts into typed evidence and writes a single JSON *envelope*; **Orbitbrief-Core** (stage two) consumes that envelope and renders the brief. A core compile never re-parses; the envelope is the sole border crossing. The flow board reproduced in Figures 1–3 is the canonical one-page depiction of stage one, and this paper is its long-form annotation.

The board is read left to right along six columns: **bytes**, **route**, **reader**, **inside that reader**, **segments**, **atoms** (left to right). Four colors carry its entire argument, and every colored box is a claim in this language:

Green — content decides. Derived from what the file is: magic bytes, structural headers, XML parts. Green things are permitted to make decisions.

Amber — name/extension: a prior only. Derived from what the file is *called*. Amber may raise a claim; it may never create one alone.

Blue — abstain: silence, recorded. The refusal states. A file nothing could read is written down as unread — a first-class outcome, not an error.

Red — where documents were being lost. Historical wounds: places the system was once wrong *silently*. Every red mark on the board is a fixed and pinned bug.

The board is self-auditing: a *board receipts* line inside its footer states that code-derived figures were re-derived on 2026-08-25 against main@f898f2e, and that corpus findings are dated measurements pinned by the audit tests that found them. Every number on the board is therefore either derived from running code or wearing its measurement date. This paper holds itself to the same standard: the header block above names the exact commits and container revisions against which every claim was verified.

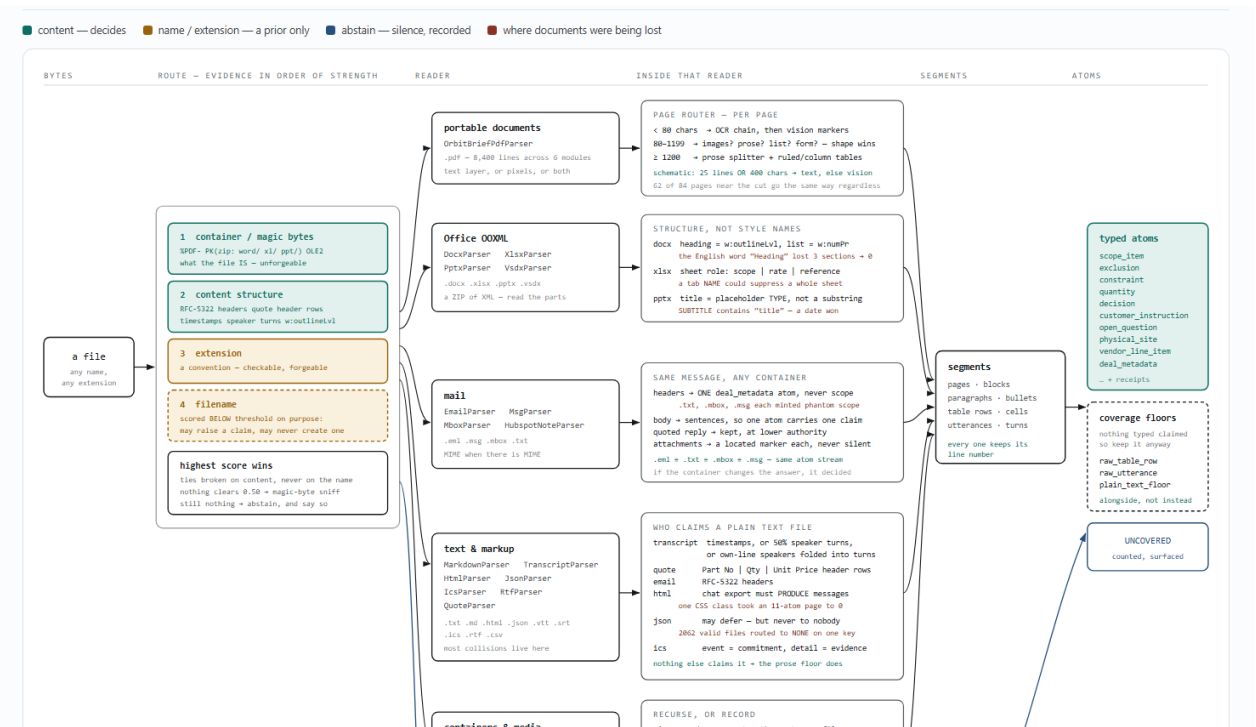


Figure 1. The board's upper region: the legend, the evidence ladder (column two), the first four reader families (column three) with their interior routing (column four), and the terminal atom vocabulary with coverage floors (column six).

2 The evidence ladder: routing by strength of evidence

2.1 Mechanism

A file arrives with *any name, any extension* — the leftmost box states the threat model outright: names lie, extensions lie, customers rename files, mail clients mangle attachments. Routing therefore descends a ladder of four evidence classes in strictly decreasing trustworthiness. **(1) Container / magic bytes** (green): %PDF-, ZIP local-file headers containing word/, x1/, or ppt/ members, OLE2 signatures — in the board's words, *what the file IS — unforgeable*: renaming a PDF cannot make its first bytes stop saying %PDF-. **(2) Content structure** (green): RFC-5322 mail headers, quote-table header rows (Part No | Qty | Unit Price), transcript timestamps and speaker turns, w:outlineLvl in OOXML — weaker than magic bytes because structure can coincide, but still content. **(3) Extension** (amber): a checkable convention, and a forgeable one. **(4) Filename** (amber, drawn with a dashed border): scored **below the routing threshold by design** — the dashed border is the architectural statement that this rung exists, contributes a prior, and is structurally incapable of deciding alone.

The *highest score wins* box completes the algorithm: ties break on content, never on the name; if no candidate clears **0.50**, a magic-byte sniff runs as a last resort; if still nothing, the file routes to the abstain path — *and says so*. The 0.50 threshold is not folklore: it carries a receipt (corpus, date, derivation, expiry) in the calibration registry (app/core/calibration.py), enforced by continuous integration, so it cannot silently expire (§13).

2.2 Models present: none — deliberately

Routing is pure rules over pure evidence. This is a defended position, not a gap. The seam-audit campaign (§10) found that across six audited layers, *zero* defects lived inside rule logic; every defect lived at a representation boundary. Rules here are cheap, inspectable, and deterministic; a learned router would add failure modes without adding capability, and the 2,500-file adversarial sweep is the empirical warrant.

2.3 Grade: A

Survived the full-corpus hijack sweep (every historical case of a filename deciding against content was found, fixed, and pinned). Thresholds receipted. The one structural risk — a new container format arriving unrecognized — degrades safely to the recorded abstain rather than to misrouting.

NOVELTY

The ladder is conventional; the dashed box is not. Encoding “this signal is deliberately sub-threshold” as a visual and architectural invariant — and generalizing it into the doctrine *a weaker signal may raise a claim it cannot create* — treats the filename as a witness who may testify but never convict. Most parsers treat filename as just another feature; the difference is precisely where all five of this system's historical routing hijacks came from.

3 The five reader families and their interiors

Twenty-one registered parsers, enumerated live from the capability table, grouped into five families. Column three names each family's parsers and extensions; column four exposes the routing *inside* the reader, which is where most of the board's red — its record of past silent losses — lives.

3.1 Portable documents: OrbitBriefPdfParser

The largest reader: approximately 8,400 lines across six modules (orbitbrief_pdf.py, pdf_image_gate.py, pdf_image_vision.py, vision_extraction.py, schematic_embedder.py, zero_miss.py — figure re-derived at the receipt date; an earlier stale figure of 12,600 was caught by the board's own receipts pass and

corrected, which is the receipts discipline working as intended). It reads the text layer, the pixels, or both. Its interior is the **page router**, a per-page trichotomy on extractable character count: **<80 characters** -> the OCR chain, then vision markers (a page that empty is an image regardless of what the file claims); **80–1,199** -> the ambiguous middle, where shape analysis decides between image-with-caption, sparse prose, list, and form; **>=1,200** -> the prose splitter plus ruled- and column-table extraction. The schematic gate — 25 text lines OR 400 characters -> treat as text, else route to vision — is receipted in the calibration registry. Beneath it sits the corpus finding *62 of 84 pages near the cut go the same way regardless* (measured 2026-08 on an 84-page probe): an honest admission that the threshold's exact placement matters less than its existence.

Models. OCR is a **Tesseract**-first chain with **EasyOCR** in cascade; rasterization via **pypdfium2**. Schematic and image pages route to a vision-language model, **Qwen2.5-VL**, which reads overlapping tiles — the overlap being why the audit's tile-deduplication fix mattered: identical bill-of-materials rows surfacing from adjacent tiles were double-counted until 2026-08-25. **GPT-4o** appears in code as a referenced alternative; the operative path is **Qwen2.5-VL** served through the **azure-ollama** proxy fleet. Schematic classification (**schematic_embedder.py**) embeds with **qwen3-embedding** when the **Mac Studio** proxy is reachable and falls back to local CPU **bge** at a receipted similarity threshold of ~ 0.62 — the fallback is a first-class path because the **Mac** is frequently offline (\$9.3).

Grade: A-. The most audited surface in the system (hyphen-wrap recall, tile dedup, and quantity-gate fixes all pinned). The minus is honest: extraction quality on dense schematics is bounded by **Qwen2.5-VL**, and the 80/1,200 cuts, though receipted, are corpus-tuned rather than learned.

3.2 Office OOXML: DocxParser, XlsxParser, PptxParser, VsdParser

The interior doctrine is its box title: **STRUCTURE, NOT STYLE NAMES**. Each line beneath is a scar with its fix. **docx**: headings from **w:outlineLvl**, lists from **w:numPr** — because trusting the English style name “Heading” once cost three sections (red: *lost 3 sections* -> 0) on a document styled in another convention. **xlsx**: sheet *role* (scope | rate | reference) is decided from content, because a tab name once suppressed an entire sheet (red). **pptx**: the title is the placeholder **TYPE**, not a substring — because a slide whose subtitle contained the word “title” once won the title slot with a date (red). A **.vsdx** is a ZIP of XML; the parser reads the parts. **Models: none** — OOXML is machine-written XML, and rules are the right tool. **Grade: A.** Every red line is historical, fixed, and pinned.

3.3 Mail: EmailParser, MsgParser, MboxParser, HubspotNoteParser

Doctrine: **SAME MESSAGE, ANY CONTAINER**. The invariant — pinned by test — is that one message yields the same atom stream whether it arrives as **.eml**, **.txt**, **.mbox**, or **.msg**: *if the container changes the answer, it decided*, and a container is amber. Headers produce exactly ONE **deal_metadata** atom and never scope (each container format once minted its own phantom scope atom — red). Bodies split to sentences so one atom carries one claim. Quoted reply chains are *kept at lower authority* (**quoted_old_email**, near the lattice floor, \$5.1) rather than deleted — demotion over deletion, always: stale evidence loses arguments instead of disappearing from them. Attachments produce a located marker each, never silently. **MIME** is honored when present. **Models: none. Grade: A.** Container-invariance is among the most elegant guarantees in the system.

3.4 Text & markup: Markdown, Transcript, Html, Json, Ics, Rtf, Quote parsers

Most collisions live here, the board admits — plain text is what every format degrades into, so the interior is a claims court: **WHO CLAIMS A PLAIN TEXT FILE**. Transcript claims on timestamps or $\geq 50\%$ speaker-turn lines, with own-line speakers folded into turns (a fixed recognition miss). Quote claims on Part No | Qty | Unit Price header rows. Email claims on **RFC-5322** headers. **HTML** must **PRODUCE** messages to claim a chat export — the rule exists because one **CSS** class once took an 11-atom page to 0 (red): presentation markup

was deciding content extraction, the single most instructive bug in the system's history. JSON may defer, but never to nobody — *2,062 valid files routed to NONE on one key* (red; the 2026-08 census that motivated the rule). ICS reads an event as a commitment with details as evidence. Beneath them all: *nothing else claims it -> the prose floor does* — the terminal guarantee that no text file falls through. **Models: none. Grade: A-** — comprehensive today, but this family is where a novel format will collide first; the minus is a standing watch, not a present defect.

3.5 Containers & media: Zip, Odt, Ods, Image, Mpp parsers

Doctrine: **RECURSE, OR RECORD**. ZIP members re-enter the router as files — full recursion, so a nested attachment receives the entire evidence ladder. ODT/ODS extract paragraphs and cells, then CLASSIFY (a default type once stamped all six atoms scope_item — red). An image with no text layer is saved and flagged for vision rather than dropped. An .mpp — a proprietary binary this system cannot open — produces a marker *so it cannot vanish*: the system records its own inability rather than pretending the file was not there. The closing line is the whole design in one sentence: *unreadable is still a file somebody must see*. **Models:** the image path feeds the same OCR/vision chain as §3.1. **Grade: A.**

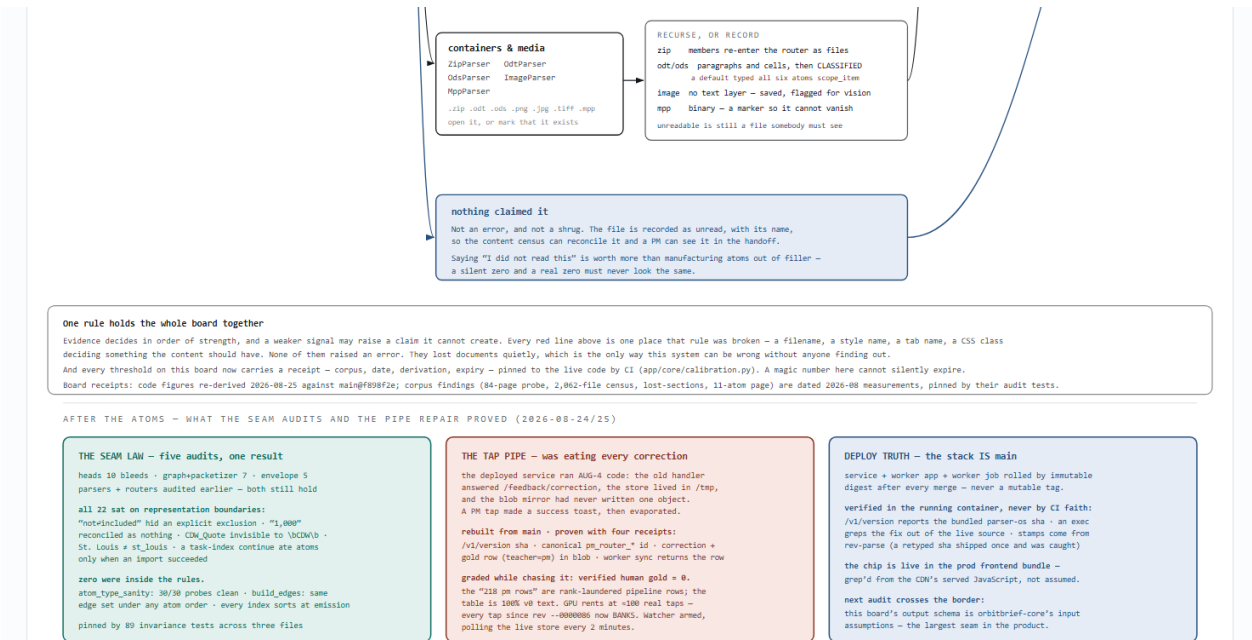


Figure 2. The board's lower region: the final reader families, the abstain box ("nothing claimed it"), and the one-rule footer with the board-receipts line.

4 Convergence: segments, typed atoms, floors, and the recorded abstain

4.1 Segments

All five families converge into one segment vocabulary — pages to blocks, paragraphs to bullets, table rows to cells, utterances to turns — and *every one keeps its line number*. This is the provenance spine: any downstream atom can point back to file, page, and line. The column's quietness is its virtue: one common intermediate representation is what lets twenty-one parsers share a single downstream pipeline. **Grade: A.**

4.2 The typed atom

The terminal green box lists the atom vocabulary: `scope_item`, `exclusion`, `constraint`, `quantity`, `decision`, `customer_instruction`, `open_question`, `physical_site`, `vendor_line_item`, `deal_metadata`, ... + *receipts*. Every EvidenceAtom carries: its type; raw and normalized text; a structured value dictionary; **entity keys** (`site:...`, `device:...`, `quantity:...` — the join keys on which the evidence graph is built, §6); source references with locators; raw confidence and **calibrated confidence**; a review status; and — the load-bearing field — an **authority class** drawn from the lattice of Table 1, which governs every downstream conflict.

rank	authority class	meaning
100	contractual_scope	the signed contract — outranks everything
95	pm_confirmed	a project manager's explicit answer — the human is IN the lattice
90	customer_current_authored	the customer's current, authored documents
~80	approved_site_roster	the approved site roster
65	vendor_quote	a vendor's quoted line items
~55	meeting_note	meeting notes and transcripts
~45	machine_extractor	machine-derived signals
~35	quoted_old_email	quoted reply chains — kept, demoted, never deleted
lowest	deleted_text	struck-through / deleted content — retained as anti-evidence

Table 1. The authority lattice. Note rank 95: a PM's answer re-enters the system as an evidence atom that outranks everything except the contract itself — which is what later makes reconciliation (§7) and instant learning (§9.1) compositional rather than special-cased.

4.3 Coverage floors — alongside, not instead

The dashed floors box lists `raw_table_row`, `raw_utterance`, `plain_text_floor`. When no typed rule claims a segment, a floor atom keeps it — *alongside, not instead*: floors never displace typed extraction; they run beneath it, so the typed layer's misses remain visible and countable rather than silently absorbed. Most systems let a fallback replace the primary path; keeping both preserves the very error signal that can later train the primary.

4.4 UNCOVERED, and “nothing claimed it”

The blue terminal boxes. Uncovered content is *counted, surfaced*. A file no reader claimed is *recorded as unread, with its name*, reconcilable against the content census and visible to the PM in the handoff. The board's own sentence is the system's ethic and deserves full quotation: “*Saying 'I did not read this' is worth more than manufacturing atoms out of filler — a silent zero and a real zero must never look the same.*”

Models in the typing layer. Rules-first, with two learned adjuncts: **SemanticRules** — embedding-prototype matching that replaces brittle regexes, offline-safe via the bge fallback — and the **calibration head**, a trained logistic model over ten signals producing calibrated confidence and the accept/needs-review verdict; it replaced a hand-tuned sigmoid that capped at 0.73 and therefore could never auto-accept anything. **Grade: A-** — the skeleton is excellent; the minus marks the acknowledged frontier: per-atom context is currently too thin for the planned neural typing heads, and typing still leans on rules awaiting learned successors that must first beat them (§9).

5 The one-rule footer and the receipts regime

The footer binds the board: “*Evidence decides in order of strength, and a weaker signal may raise a claim it cannot create. Every red line above is one place that rule was broken — a filename, a style name, a tab name, a CSS class deciding something the content should have. None of them raised an error. They lost documents*”

quietly, which is the only way this system can be wrong without anyone finding out.” Two mechanisms make this more than rhetoric. The **calibration registry** pins every threshold on the board — corpus, date, derivation, expiry — enforced by CI, so a magic number cannot silently expire. The **board receipts line** extends the same discipline to the diagram itself: code figures re-derived against a named commit, corpus findings wearing their measurement dates. **Grade: A+**, earned on 2026-08-25 when the receipts pass caught the board's own stale line-count figure — the discipline catching its own author.

6 Beyond the board, I: the evidence graph and the packetizer

Atoms are vertices; `build_edges` (in `graph_builder.py`, ~1,550 lines) connects them across artifacts on shared entity keys, emitting typed edges — **supports**, **contradicts**, **excludes**, **requires** — under named edge families (value support, constraint alignment, quantity contradiction, part-number quantity conflict, exclusion application, device/material aggregate mismatch, cross-artifact co-mention, schematic quantity contradiction, and others). Guardrails bound the construction: candidate-pair caps, per-atom pair caps, noisy entity-key floors, and email-header suppression (a To:/From: line mis-typed as an exclusion once flooded the graph with excludes edges). Quantity binding is noun-anchored — an integer binds to a device noun only within a tight same-clause window (30 characters before the noun, 15 after), with sentence boundaries terminating the window so “Install 12 APs. We will install 4 switches” cannot bind 12 to switches. The audit proved and pinned the property that matters most: **the edge set is invariant under input atom order** (verified on a non-vacuous 11-edge fixture across shuffles; the pinning test asserts its own fixture is non-vacuous so it cannot rot into a trivial pass).

The **packetizer** (~1,930 lines) assembles certified **evidence packets** — the PM-facing unit of “here is a real issue, with receipts.” Certification is conservative by construction: a `quantity_conflict` packet requires at least two quantity atoms, at least one contradicts edge between them, AND multi-source provenance (two artifacts or two authority classes) — *the system never certifies a single source arguing with itself*. A narrow, receipted exception admits same-sheet schematic contradictions only when both sides carry replayable pixel-region provenance. Packet families cover quantity conflicts, scope exclusions, site access, certification requirements, pathway/raceway uncertainty, procurement status, and more; each family has authority allowlists for what may govern it. **Models: none** in graph or packetizer — these layers are deliberately symbolic, because their outputs must be defensible line by line to a human. **Grade: A** for both, post-audit (the “not\nincluded” line-wrap fix and comma-quantity fixes are pinned).

7 Beyond the board, II: phase-2 reconciliation

Where the graph finds contradictions and the packetizer certifies them, **reconciliation** (`reconcile.py`) adjudicates them. Conflicting claims about the same scope key form a conflict set; the authority lattice (Table 1) resolves it when one claim strictly outranks — producing a **verdict with receipts**: the winning value, the winner's authority and rank, every superseded claim with its authority and rank, the edge identifiers that proposed the conflict, and a human-readable reason of the form “56 governs because *customer_current_authored (rank 90) beats '40' at rank 65*.” When the top ranks tie, the system **refuses to guess**: the conflict is emitted as open, with both claims and the reason “*picking between equals is a PM's judgment*.” The resolver is an injectable **Potential** — a function from a conflict set's claims to a winner index or an abstain — with the authority lattice as the default; a learned potential slots in per conflict-set *when a corpus earns it*, and the honest-abstain path survives either way. Every emitted list travels with its precision context: the rule that *proposes* contradictions measured **41% precise on 447 labelled pairs** (supports: 90%), so each entry is presented as *a lead with receipts, not a verdict without them* — the caveat is embedded in the data structure itself and rendered into the brief (§8). **Grade: A-** — architecture and honesty are exemplary; the minus is that the default potential is still the hand-built lattice, pending the learned successor and, before that, a labelled

corpus that is not circular (§9.4).

NOVELTY

The lattice-with-injectable-potential is the system's neuro-symbolic thesis in one mechanism: symbolic authority as the default judge, a learned model admitted seat-by-seat only on demonstrated evidence, and abstention preserved as an outcome under either regime. Equally uncommon: the measured precision of the conflict-proposing rule travels WITH the conflicts into the human-facing document, so the reader's trust is calibrated by the pipeline itself.

8 Beyond the board, III: the envelope, the border, and the brief

Stage one's terminal act is the **envelope** (`orbitbrief_envelope.py`, ~1,930 lines): the `orbitbrief.input.v2` JSON written to Azure blob under the deal's latest compile. It carries the documents (with per-artifact parse outcomes), the compacted atoms (type, authority, text, section path, locator, entity keys, structured values, calibrated confidence, verification state), entities, edges, packets, drawings, summary counts, sorted indexes (by section path, type, authority, artifact, entity key, stakeholder, device, site — all emitted pre-sorted, hence order-invariant), the deal-financials block, service routing, and the phase-2 reconciliation key. Envelope-level defenses include the **foreign-artifact check** — a filename claiming a distant deal number is flagged (never dropped: a related document is sometimes filed on purpose, and deleting evidence on a filename heuristic would itself violate the ladder), with adjacent numbers tolerated because 60 of 79 observed mismatches were benign siblings and flagging them would have made the signal 76% false; the check exists because five artifacts from three other opportunities once produced a fluent, confident brief about the wrong company. *Incomplete is recoverable; confidently wrong is not.*

The **border**: Orbitbrief-Core validates the envelope with a lenient Pydantic model (`EnvelopeV2`, `extra="allow"`) so the producer may ship new keys first — the audit pinned a round-trip test proving novel producer keys survive rather than being stripped or rejected. Core's consumers then project the envelope into the **PM handoff**: domains led by the routed service, gap cards, risk register, schedule, site rollups, money and date mentions, executive summary — and, since 2026-08-25, the reconciliation section: “*Cross-document conflicts — resolved by authority, and the ones that need you*”, open ties first, receipts attached, the 41% caveat in-line; envelopes predating the key render nothing. One dataclass field feeds all three surfaces (JSON, Markdown, HTML — the HTML wraps the Markdown). The brief's narrative prose is generated by **Qwen3-14B served via Ollama** through the azure-ollama proxy fleet (with `/no_think` and stream passthrough); retrieval and semantic features use **Qwen3-Embedding-8B** / Azure OpenAI embeddings with the same bge fallback ethic. **Grade: A** — the border is tolerant, tested, and now consumes every major envelope surface; the minus: the human-readable contract file lags the model (cosmetic), and the reconciliation section awaits its first real conflict deal to debut.

9 The learning loop, the model inventory, and where the GPU is

9.1 The correction loop, end to end

A **CorrectionChip** in the production frontend lets a PM re-route a deal's detected service in two taps. The tap posts to `/feedback/correction` on the parser-os service, which mints a canonical correction (deal-scoped, exemplar text captured) into the FeedbackStore and — durably — into a blob mirror: the correction object plus a **teacher=pm training row**. Workers rehydrate the store from blob on every compile, so the correction **governs the very next compile of that deal** — instant learning, no retraining. The same rows accumulate as the gold corpus for the GPU fine-tune (§9.3). Two timescales of learning from one human gesture, and the

gold-collection instrument is the product's own correction UI rather than a separate labelling operation.

The loop's history is the band's red panel and is retold here without euphemism: until 2026-08-25 the deployed service was running August-4 code — an obsolete handler answered the route, the store lived in ephemeral /tmp, and the blob mirror had never written a single object. A PM's tap produced a success toast and evaporated. The repair was rebuilt from main, rolled by digest, and proven with four receipts: the version endpoint's source sha, the canonical correction id, both blob objects, and an in-container worker sync returning the row. A separate grading pass established the harder truth: the training table's “218 PM rows” were rank-laundered pipeline rows with empty provenance — **verified human gold = 0**. Every tap since service revision 0000086 banks durably; a watcher polls the live store on a two-minute cycle and will flag the first row whose author contains an @ — the first real human gold in the system's life.

9.2 Complete model inventory, by residence

model	role in the pipeline	where it runs
Tesseract + EasyOCR	OCR chain for image pages (§3.1)	CPU, inside the parse workers (Azure Container Apps)
Qwen2.5-VL	vision extraction on schematic/image tiles	azure-ollama proxy fleet
DeepSeek (deepseek-chat)	zero-miss escalation tail; teacher prefill; rubric relabels; question-quality audits	external API
Qwen3-14B (/no_think)	brief narration in Orbitbrief-Core	Ollama, behind azure-ollama proxies
qwen3-embedding / Qwen3-Embedding-8B	SemanticRules, schematic classification, retrieval	Mac Studio (via Tailscale + proxies) — frequently offline, by tolerated design
bge (base/small)	embedding fallback, threshold ~0.62	CPU, in-container — the always-available path
calibration head (logistic, 10 signals)	calibrated confidence + accept verdict	CPU everywhere (trained, shipped)
service-router head (kNN over embeddings)	deal-level service routing with abstain	CPU in-container; its contrastive fine-tune is THE GPU job
GNN v2 (relation-edge)	edge classification (rules 0.49 -> 0.99 on labels)	standalone — not yet in production

Table 2. Every model in the system and its residence. Note the absence of any GPU row on the serving side.

9.3 Where the GPU is — the precise answer

There is no GPU anywhere in the serving path. Every deployed container — service, both parse workers, the brief worker — is CPU-only on Azure Container Apps. The GPU in this system's life is a **rented A100**, provisioned on demand exclusively for training, and it lives on **neither Azure's serving plane nor the Mac**. Its input is already staged on Azure blob: the leak-audited 103-deal enriched trainset (_training_deepseek_gpu.db) and the 95.9%-keep clean-rubric relabel. The Mac Studio's only role is inference — optional embedding service — and every consumer of it carries the bge fallback, so the system tolerates its absence by design. The economics are explicit and honest: the A100 rent becomes worth paying at approximately **100 real PM taps**; the current verified count is **zero**; therefore the entire training apparatus — the one-command GO#5 trainer with six pinned decisions, proven end-to-end on synthetic gold, and the eval-gate that requires beats-baseline before any promotion — is loaded, tested, and deliberately idle. Nothing trains on fake gold; nothing promotes on hope.

9.4 Grades for the learning system

Tap pipe: **A** (proven end to end, durably banked, watched). Trainer and eval-gate: **A-** (proven on synthetic gold; unproven on real gold because none exists). Learned-model maturity overall: **B-**, and the grade is a policy

statement rather than a defect: the calibration head is trained and shipped, but the router head fails its own eval-gate pending the GPU fine-tune; router gold labels are circular (a DeepSeek grader reproducing DeepSeek labels, 91% self-agreement — no trustworthy router accuracy exists yet); the deployed router consequently routes to only ~4 of the 29 recognized workstreams, though the correction vocabulary spans all 29 (verified: all 29 labels store as corrections). A system that refuses to promote models on circular gold earns a B- exactly this way.



Figure 3. The appended band: the seam law (green), the tap-pipe repair (red — the board's color for silent loss, which this was), and deploy truth (blue), with the campaign's counts and receipts.

10 The seam-audit campaign and its law

Six audits — parsers, routers, post-parse heads, graph+packetizer, envelope, and the cross-repository border — under one falsifiable method: **hold the content constant, vary the incidental dress** (casing, punctuation, whitespace, line wraps, thousands separators, input order, even import success), and count what changes that should not. The result, stated as the campaign's law: **twenty-two defects in the final four audits, every one at a representation boundary — two encodings of the same truth disagreeing — and zero inside rule logic.** Table 3 catalogues the instructive exhibits.

defect	boundary	consequence until fixed
"not\nincluded" invisible	PDF layout / phrase	a line wrap hid an EXPLICIT EXCLUSION from packet governor logic — the highest-stakes phrase family in the product, defeated by where a line broke
"1,000" bound nothing	display format / number	the digit regex split the token; separately, string "1,000" parsed to None — comma-formatted quantities silently exited cross-document reconciliation
covered-check never matched	de-spaced root / spaced text	every covered multi-word PM-critical term re-escalated to the paid LLM on every deal — invisible duplicate spend
St. Louis vs st_louis	text / slug	a period broke site-catalog matching in both directions
CDW_Quote invisible	separator / word boundary	underscore is a word character, so \bCDW\b could never see the account name — degrading the misfiled-document signal built after the wrong-company brief
hyphen-wrap blind sweep	PDF hyphenation / term	"perfor\nmance bond" counted zero mentions — the recall net skipped exactly the tight-layout documents it exists for
the task-index continue	import success / control flow	non-quote-line task atoms vanished from every entity index — and only when a classifier import succeeded, so environments could disagree about envelope contents
"48,500.00" erased the total	JSON dressing / value	a string-typed revenue made the brief's headline deal total None (the border audit's fix, in core)

defect	boundary	consequence until fixed
tile-duplicate rows	tile / tile	the same BOM row from overlapping vision tiles survived dedup twice and double-counted
quantity keys unguarded	emitter path / emitter path	Qty: 0 and 2,000,000 emitted entity keys through two of three paths, polluting rollups — the plausibility gate covered only the third

Table 3. Selected exhibits from the 22 audited defects. Complete listings live in the five pinning test files (Appendix C).

Equally important is what **held**, now pinned so it keeps holding: `atom_type_sanity` survived all 30 of its probes untouched; `build_edges` emits an identical edge set under any atom input order; material keys collapse “Cat6A”/“Cat 6A”/“CAT6A patch”/“cat-6a” to one identity; every envelope index sorts at emission; core's 29 domain packs and the frontend's 29 workstreams agree exactly; all 29 correction labels store end-to-end. Roughly **114 invariance tests** across five files in two repositories pin both the fixes and the held properties — with the order-invariance test asserting its own fixture is non-vacuous, so the pin cannot rot into a trivial pass. **Grade: A+.**

NOVELTY

The law itself is the exportable finding: in a mature evidence-processing system, hand-written rules were blameless across six audits, while every defect was two representations of one truth disagreeing — slug versus text, layout versus phrase, format versus value, import success versus control flow. The corollary is a portable audit method: probe the seams first, with dress-invariance as the falsifiable property, and pin what holds as firmly as what broke.

11 Deployment and verification discipline

Three rules, each purchased with a specific failure. **(1) Stamps from rev-parse, never retyped:** images are tagged by source sha with the provenance stamp taken programmatically — a hand-retyped sha (off by one digit, naming a nonexistent commit) shipped exactly once and was caught by the verification step it exists to serve. **(2) Roll by immutable digest, never mutable tag:** a mutable-tag roll once shipped week-old code silently while claiming success. **(3) Verify in the running container, never by CI faith:** after every roll, the version endpoint must report the expected source sha and an exec must grep the actual fix out of the live container's source (or import the new symbols and execute the fixed function). The current verified state is the header block of this paper. **Grade: A+.**

12 Where the system currently stands

Everything described in this paper is merged, deployed, and container-verified as of the header date. The pipeline is audited end to end under a falsifiable method; both visualizations agree with the code and carry their own receipts; the correction loop is proven with synthetic gold and watched continuously. The single limiting reagent is human: **zero verified PM taps**. The activation sequence is fully determined: at the first tap, the row banks durably and governs that deal's next compile within minutes; at approximately one hundred taps, the rented A100 becomes economically justified and the router's contrastive fine-tune unlocks; the eval-gate then decides — on beats-baseline evidence, never on hope — whether the learned head earns promotion; and the injectable-potential architecture (§7) extends the same earned-seat offer to reconciliation. The machine is built, measured, and idling at the line with the engine proven. What it awaits is a thumb.

region	grade	one-line justification
Evidence ladder + routing	A	hijack-sweep clean; thresholds receipted; filename provably sub-threshold

region	grade	one-line justification
PDF reader + page router	A-	most-audited surface; vision bounded by Qwen2.5-VL; cuts receipted but corpus-tuned
OOXML reader	A	structure-not-style; every scar pinned
Mail reader	A	container-invariance guaranteed by test
Text & markup claims court	A-	comprehensive; first collision zone for novel formats
Containers & media	A	recurse-or-record; nothing can vanish
Segments	A	one vocabulary, full provenance
Atoms + floors + abstain	A-	excellent skeleton; context enrichment and learned typing pending
Footer + receipts regime	A+	self-auditing board; CI-pinned thresholds
Evidence graph + packetizer	A	order-invariant; conservative certification; audited
Phase-2 reconciliation	A-	verdicts with receipts; honest abstain; learned potential pending a non-circular corpus
Envelope + border	A-	tolerant, tested; contract doc lags model (cosmetic)
Tap pipe + instant learning	A	proven with four receipts; watched; awaiting humans
Deploy discipline	A+	digest rolls; in-container proof; rev-parse stamps
Seam-audit campaign	A+	6-for-6; ~114 pins; falsifiable method; exportable law
Learned-model maturity	B-	calibration head shipped; router awaits GPU fine-tune; zero verified human gold; no promotion without beating baseline — by policy

Table 4. The consolidated grade sheet. Every grade is argued in its section; disagreement with a grade is a conversation the receipts are designed to enable.

13 The genuinely novel, collected

1. The abstain economy. Refusal as a first-class, PM-visible product surface at every layer: routing abstains and says so; floors count the typed layer's misses; reconciliation refuses top-tier ties; the router head abstains below threshold; an unreadable file becomes a marker somebody must see. The system's most consistent design choice is knowing what it doesn't know, out loud.

2. The receipts regime. Thresholds carry corpus/date/derivation/expiry enforced by CI; deploys are proven in the running container; conflict lists carry their proposing rule's measured precision; the diagram of the system carries its own re-derivation stamp. “Trust me” appears nowhere in the architecture.

3. The seam law. An empirical, falsifiable finding about where defects live in evidence-processing systems — at representation boundaries, not inside rules — discovered by a repeatable dress-invariance method and enforced by ~114 pins.

4. The authority lattice as neuro-symbolic spine. One total order performs conflict resolution, quote demotion, PM-answer integration (rank 95), and reconciliation verdicts — with an injectable potential so a learned model can take the judge's seat per conflict-set when a corpus earns it, and honest abstention preserved under either regime.

5. Instant learning without retraining. A PM's tap governs the next compile via store rehydration while the same row accumulates toward the GPU fine-tune — two learning timescales from one gesture, with gold collection embedded in the product's own correction UI.

6. A weaker signal may raise a claim it cannot create. The ladder doctrine, generalized system-wide: filename below threshold, containers that must not change the answer, presentation markup that must produce content to claim it. Every red mark on the board is a place this rule was once broken; none of them is current.

Appendix A Deployment receipts (verified in-container, 2026-08-25)

target	revision	image digest (prefix)	runs
parser-os-service-dev-eus2	0000088	38f2962c	parser-os main@f898f2e
parser-os-worker-warm	0000223	512f8f6f	parser-os main@f898f2e
parser-os-worker-dev-eus2 (job)	template	512f8f6f	parser-os main@f898f2e
orbitbrief-core-worker-dev-eus2	0000117	4ce4485d	core feat/neural-heads@914763c (bundles parser-os main)

parser-os main HEAD at writing is f523475, one docs-only commit past the deployed f898f2e — deployed code equals main code; only documentation differs. Verification methods: the /v1/version endpoint reports the bundled source sha; container execs grep fixes from live source and execute fixed functions (e.g., the string-money coercion returning 48500.0 inside the running brief worker).

Appendix B The corpus findings and their dates

finding	measured	meaning
84-page router probe: 62 near-cut pages route identically either side of the schematic threshold	2026-08	the threshold's placement matters less than its existence
2,062 valid JSON files routed to NONE on one key	2026-08	the census behind “may defer — but never to nobody”
one CSS class took an 11-atom page to 0	2026-08 (archaeology)	presentation markup deciding content — the claims-court rule's origin
“Heading” style-name trust lost 3 sections	2026-08 (archaeology)	the origin of structure-not-style
contradicts-rule precision 41% on 447 labelled pairs (supports 90%)	2026-08	the caveat that travels with every conflict list
60 of 79 deal-number mismatches were adjacent siblings	2026-08	why the foreign-artifact check tolerates adjacency (else 76% false)
router gold self-agreement 91% (DeepSeek grading DeepSeek)	2026-08	why no trustworthy router accuracy exists yet

Appendix C The pinning test files (the audit record)

file	tests	pins
parser-os/tests/test_head_invariance.py	43	the 10 head bleeds + held head invariants (site periods, sweep root, hyphen wrap, quantity gates, tile dedup)
parser-os/tests/test_graph_packet_invariance.py	35	comma quantities, exclusion whitespace, edge-set order invariance (non-vacuity asserted), collapse determinism, gate dress
parser-os/tests/test_envelope_invariance.py	11	separator word-boundaries, foreign-artifact dress, task-index completeness, index order invariance
Orbitbrief-Core/tests/test_border_invariance.py	18	money in every dress, router-reader invariants, new-producer-key round-trip survival
Orbitbrief-Core/tests/test_reconciliation_verdicts.py	7	verdict projection, caps-vs-uncapped counts, old-envelope no-op, receipts + caveat in rendered output

Appendix D Glossary

abstain — a recorded refusal to decide — routing abstains on weak evidence; reconciliation abstains on top-tier ties; the router head abstains below confidence threshold. Always recorded, never silent.

atom (EvidenceAtom) — the unit of extracted evidence: one typed claim with text, structured value, entity keys, authority class, confidence, and source references.

authority class / lattice — the total order of Table 1 governing which evidence wins a conflict; the human PM sits inside it at rank 95.

bleed — the campaign's term for a defect where incidental dress changed an outcome that content alone should determine.

calibration registry — `app/core/calibration.py` — the CI-enforced ledger where every threshold carries corpus, date, derivation, and expiry.

coverage floor — an untyped catch atom (`raw_table_row`, `raw_utterance`, `plain_text_floor`) that keeps content no typed rule claimed — alongside, never instead of, typed extraction.

digest roll — deploying a container by its immutable image digest rather than a mutable tag, so what runs is provably what was built.

dress — incidental representation: casing, punctuation, whitespace, line wraps, thousands separators, input order. The audit method varies dress while holding content constant.

edge — a typed relation (supports / contradicts / excludes / requires) between two atoms, proposed by rules with measured precision.

entity key — a normalized join key on an atom (`site:...`, `device:...`, `quantity:...`) on which the evidence graph connects atoms across artifacts.

envelope — the `orbitbrief.input.v2` JSON — stage one's complete output and the sole border crossing into stage two.

eval-gate — the promotion rule: a learned head ships only when it beats the incumbent baseline on held-out labels.

gold — a training label whose provenance is a verified human action. Current verified count: zero — the system's honesty about this is load-bearing.

instant learning — a correction governing the next compile via store rehydration, with no retraining.

packet — a certified, PM-facing bundle of atoms + edges asserting one issue, with multi-source provenance requirements.

pin — an invariance test that fixes a property (or a fixed defect) in place so regression is impossible without a red suite.

potential — the injectable resolver in reconciliation: `claims -> winner-or-abstain`; the lattice by default, a learned model when earned.

receipt — the evidence attached to a claim — a threshold's derivation, a verdict's superseded claims, a deploy's in-container proof.

seam — a representation boundary where the same truth is encoded twice (`slug/text`, `layout/phrase`, `format/value`) — where all bugs lived.

tap — a PM's two-gesture correction in the product UI; one tap = one durable correction + one `teacher=pm` training row.

teacher=pm — the provenance tag marking a training row as sourced from a verified human correction.

verdict — reconciliation's resolved conflict: winner, superseded claims, ranks, edge ids, and a human-readable reason.

References: the system itself. `docs/parser_flow.html` (the board; Figures 1–3 are its light-theme render), `docs/parse_map.html` (the deep narrative map), `docs/AUDIT_HANDOFF.md` (the independent-audit mission brief), and the five pinning test files of Appendix C, which constitute the campaign's primary record. Model inventory derived by `grep` over both repositories on 2026-08-25; deployment receipts verified against running containers the same day.